



World's First

ENIAC was the world's first supercomputer. It weighed in at nearly 3 tons, used 19,000 vacuum tubes and took up a whopping 15,000 square feet of floor space

Feature

such as John the Ripper and Cilia can be used to crack passwords much quicker than a single computer. Check out <http://bit.ly/b4zvFI> for complete documentation on password security.

3.Cure cancer

You may want to consider joining the Folding@home project which is a distributed computing project run by the good folk at Stanford University and managed by Professor Vijay Pande. Their aim is to understand protein-folding, which is related to diseases such as Alzheimer's, Huntington's, Parkinson's and many cancers. All you have to do is download and install the software, and contribute your computer resources. Go to <http://folding.stanford.edu> for more details.

4.Little green men

If you believe that life exists out there, you can help scan all the radio noise in space by dedicating computing power to finding broadcasts from other planets. The Search for Extra-Terrestrial Intelligence is what you can contribute to, by visiting their SETI@home project at <http://setiathome.berkeley.edu/>.



To boldly go where no man has gone before!

5.Video transcoding

Transcoding HD (Blu-ray) video takes up to seven times longer than SD video. With open source tools such as the Media Encoding Cluster Project (<http://www.codegrid.de/trac>), you can effectively utilise your Windows or Linux cluster to rip your favourite flicks.

General cluster?

A cluster is essentially a group of computers linked together and performing their duties as one powerful (and power hungry) computer. The computers are usually linked using

Light speed - World's top 3 supercomputers

#3 IBM BladeCenter QS22/LS21 "Roadrunner", country : USA

Clocking in at 1.04 petaflops per second, this former world no.1 occupies a staggering 6,000 sq. ft! When compared to the 100 sq. ft. per capita space that the average Indian has, this monstrosity simulates the ageing of nuclear materials to check the viability of USA's nuclear arsenal.

#2 Dawning TC3600 Blade "Nebulae", country : China

This 1.27 petaflops per second beast is theoretically capable of running at almost 3 petaflops per second, the fastest ever!

#1 Cray XT5 "Jaguar", country : USA

As of May 2010, the Cray Jaguar is the fastest supercomputer in the world. Delivering a knock out punch to the IBM Roadrunner's 18 month tyranny as world's no.1, this 1.75 petaflops per second supercomputer boasts 2,24,000 cores which would make it the equivalent of 56,000 quad core PC's!

standard Ethernet cables, and it's obvious that Gigabit networks are preferred.

However, if you think you can do anything on a cluster that you can on a PC, you're going to be disappointed. It's a software problem. Most software we use these days are not even optimised to run on multiple CPU core PCs, let alone a cluster of multiple PCs. We can hope that this will change soon, what with everyone leaning towards multi-core computing.

Grease up those elbows

Now it's time to start building your cluster. A cluster can be architecture independent as long as all the computers on the cluster run the same SSI (Single System Image). So even if you have PCs with different operating systems and hardware configurations, a cluster centric Linux distribution such as CHAOS, ROCKS or PelicanHPC can be booted into memory of a server, and then all the others can do a LAN boot, using the SSI from the primary. The primary is the one that splits up tasks into chunks and offsets it to the nodes to be computed, and then re-assembles it all and displays the results. Thus, it's best to use the most powerful

computer on the grid as the primary to prevent even tiny amounts of latency.

Although it would be quite possible to build a cluster composed of both Linux and Windows PCs, for the purpose of this article, we'll be using a Linux distro called PelicanHPC (High Performance Cluster). The latest release of PelicanHPC is on the Mindware DVD, along with a PDF document that has loads of advanced information – how to save your work, how to create customised PelicanHPCs, etc.

Getting started

PelicanHPC is a Live CD image that allows you to get a cluster up and running in a matter of minutes. It allows you to do parallel computing using what's known as a Message Passing Interface (MPI). It's this MPI that makes it possible to split tasks up into smaller sub-tasks which are transmitted to and fro as "messages". You can run Pelican on a single, multiple-core machine, or just network as many PCs as you can together to make a cluster.

While it's possible to boot the primary system from a pen drive, we strongly recommend burning the ISO to a CD and using that instead. Since it's a Live CD, you don't have to worry about formatting or rearranging your parti-

HPC Test	-----
Quantity of processors	= 19
Calculation time	= 0.28 seconds
Cluster speed	= 6483 MFLOPS

'Digit'ised cluster : An example of cluster scalability.

tions, and when PelicanHPC is turned off, all the PCs go back to their original states. Of course you can also choose to install PelicanHPC on the primary's hard drive which would allow you to always continue where you left off from.

Requirements

Obviously, all PCs that are part of the cluster should be networked. Use a

India's top number crunchers

- #3 IBM eServer BlueGene/L, location: IISC, Bangalore
- #2 PARAM cluster, location: C-DAC, Pune
- #1 HP Cluster Platform 3000, location: Computational Research Laboratories, Pune.